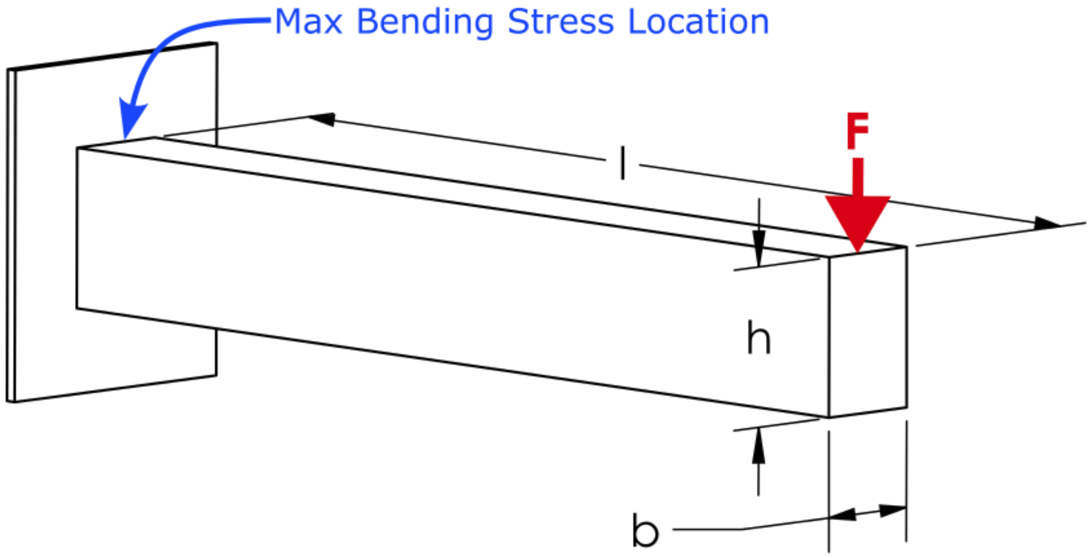
Database Consistency Test

Introduction

This EngineeringPaper.xyz sheet shows how to calculated the maximum stress and displacement for a cantilever beam loaded by a vertical force at its end. The geometry of the beam is shown in the figure below. The beam has a rectangular cross section with height h and width b. The maximum stress in a cantilever with length much larger than its height is due to bending stress that occurs at the base of the beam. For a downward force, the maximum stress will be a tensile force at the top of the beam and a compressive stress at the bottom of the beam. For beam sections that are symmetric top to bottom, the maximum tensile and compressive stresses will have equal magnitude. µ ρ 你好吗？请问你想把“cats and dogs”翻译成普通话吗？ 🤣😂😊🌴🌳🌲 #



Select Beam Material

$$\text{Aluminum Alloys} \begin{cases} E = & 71.7[GPa] \\ v = & 0.34 \\ \rho = & 2800\left[\frac{kg}{m^3}\right] \end{cases}$$

Source: Norton, R. L. “Machine design. A integrated approach, 5th Editi.” (2013).

Select Beam Section

$$\text{Rectangle} \quad \begin{cases} I_x &= \frac{b \cdot h^3}{12} \\ c &= \frac{h}{2} \\ r &= [mm] \\ r_1 &= [mm] \\ r_2 &= [mm] \\ a &= [mm] \\ b &= 50[mm] \\ h &= 75[mm] \\ b_1 &= [mm] \\ h_1 &= [mm] \end{cases}$$

Source for beam section equations and images: https://en.wikipedia.org/wiki/List_of_second_moments_of_area

Define the Other Beam Parameters

$$l = 500[mm]$$

$$F = 500[N]$$

Define the Deflection and Stress Equations

The maximum displacement of a cantilever beam loaded at its end is defined by [1]:

$$y_{max} = -\frac{F \cdot l^3}{3 \cdot E \cdot I_x}$$

where Ix is the area moment of inertia of the cross section about the x-axis as selected in the table above. The maximum displacement can now be queried:

$$y_{max} = [mm] = -0.165297794307557[mm]$$

The maximum bending stress in a beam section is given by:

$$\sigma_{max} = \frac{M \cdot c}{I_x}$$

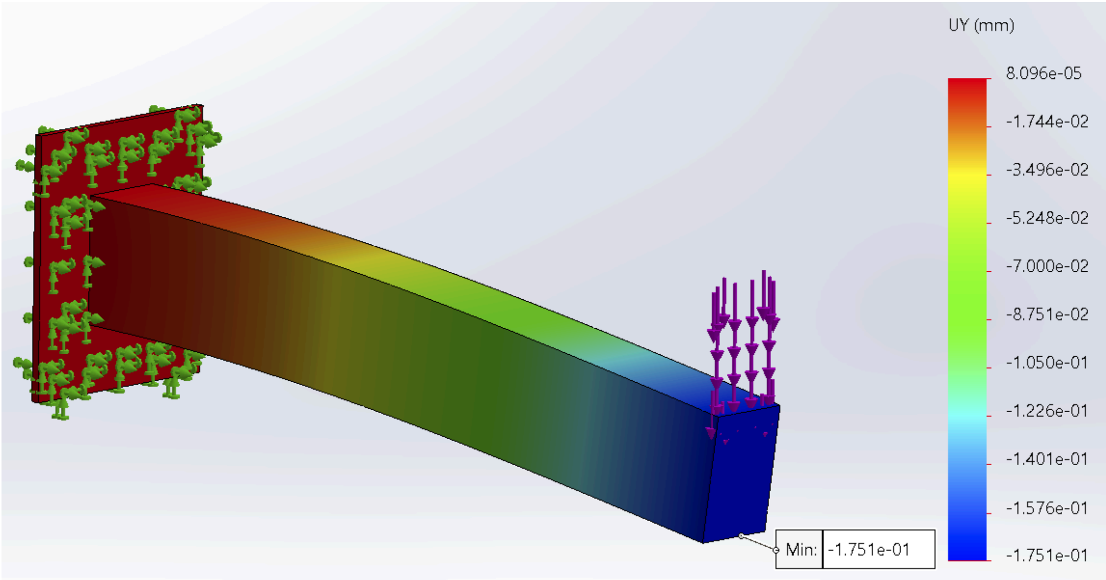
where M is the bending moment at the section of interest and c is distance from the neutral axis of the beam to the top or bottom of the beam. The maximum moment occurs at the base of the beam and is defined by:

$$M = l \cdot F$$

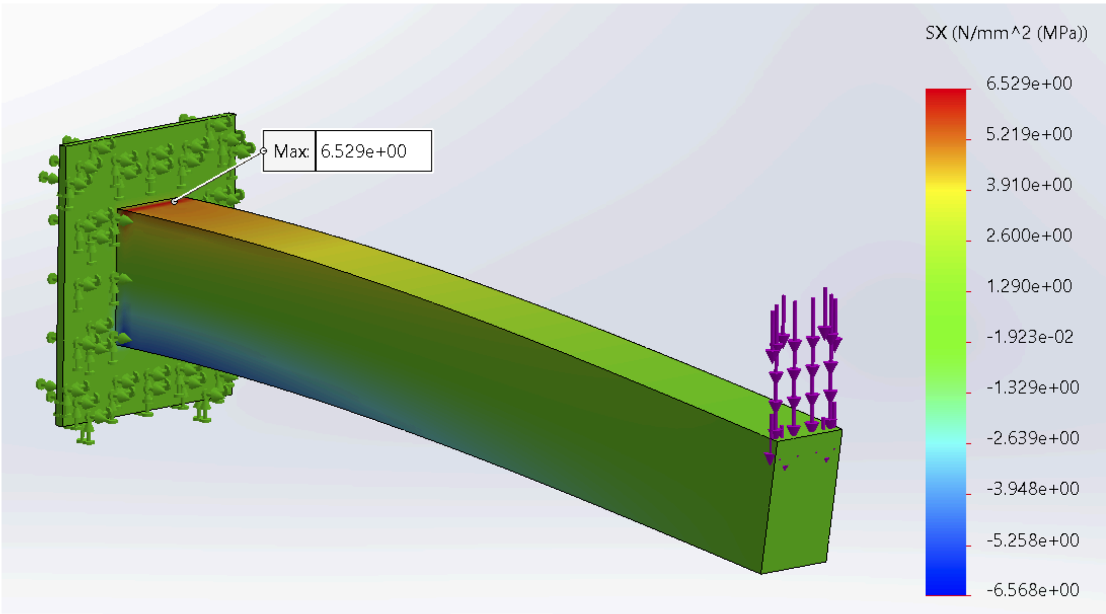
where l is the length of the beam. Now that everything is defined, the maximum stress can be queried and converted to [MPa] units:

$$\sigma_{max} = [MPa] = 5.33333333333333[MPa]$$

These results can be compared to the results obtained from a finite element analysis using SolidWorks. The beam has the same dimensions, loading, and material as the rectangular cross-section and the aluminum table options above. The displacement plot is shown below. The maximum displacement magnitude obtained using finite element analysis is .175 [mm], which is close to the .165 [mm] obtained using the equations above.

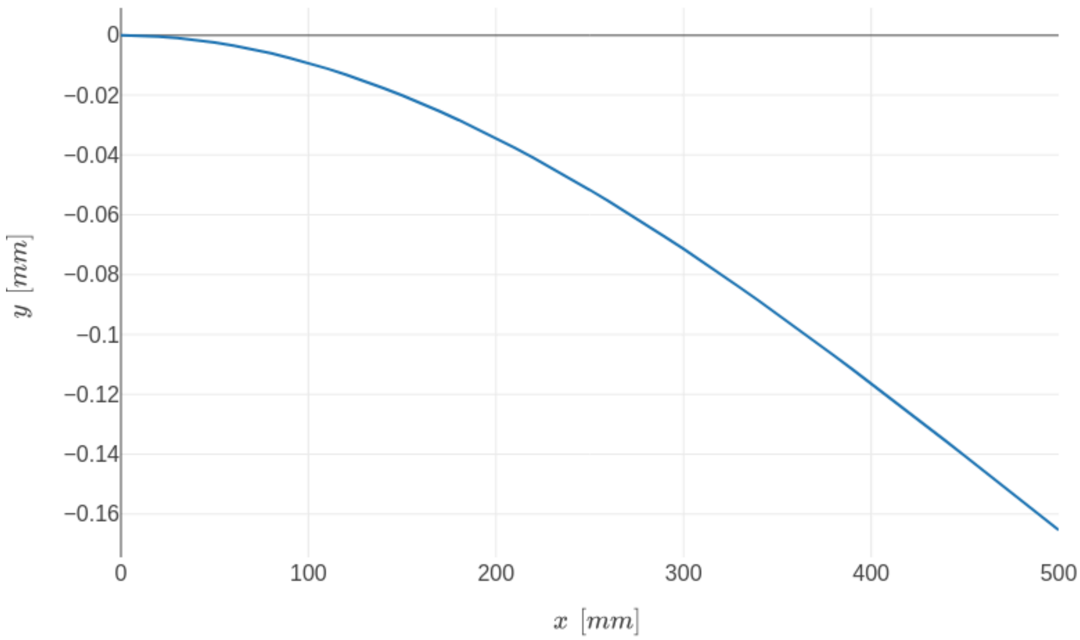


Similarly, the maximum stress calculation can be compared to the SolidWorks results. The maximum stress obtained from SolidWorks is 6.5 [MPa] as shown below. Note that this is higher than the 5.3 [MPa] obtained using the beam equation. This is the result of the stress concentration that occurs at the based of the cantilever where it meets the fixed boundary condition. A littler further from the fixed boundary condition, the finite element values are close to the calculated values.



[1] Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

$$y = \frac{F}{6 \cdot E \cdot I_x} \cdot (x^3 - 3 \cdot l \cdot x^2)$$



All Colors

one

two

three

four

five

six

seven

nine

ten

eleven

twelve

thirteen

fourteen

fifteen

sixteen

seventeen

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MD Export Test (centered equations unchecked)

$x = 20[m]$

Documentation cell with line of text followed by a scaled image:

Sentence with formula $\sqrt{\Omega}$ in the middle, followed by equation on its own line:

$E = \frac{My}{I}$

With text after.

Line of text.

Another line of text.

Option 2 $\begin{cases} l_1 = & 3[mm] \\ l_2 = & 4 \end{cases}$

$l_2 = 4$

$l_1 = [m] = 0.003[m]$

$Col1$	$Col2$
	$[m]$
6	7
8	9

$y = \begin{cases} 10: & x>20 \\ 2\ : & \text{otherwise} \end{cases}$

System $= \begin{cases} 1 \cdot s + 2 \cdot t = 10 \\ 2 \cdot s - 4 \cdot t^2 = 20 \end{cases}$

Solution $= \begin{cases} s = & 12 \\ t = & -1 \end{cases}$

Roller Thrust Bearing $\mu_c = 0.01$

This left justified, below is a centered equation:

$$x = 2 \cdot y$$

Below is a centered image:

This is right justified

This is justified text

- red
- blue

1. one
2. two

    a. three
    b. four

Normal text

MD Export Test (centered equations checked)

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Another line of text.

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$Col1$	$Col2$
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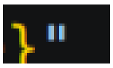
Normal text

Equation Comments Stress Test

cats and dogs

cats and dogs

@cats



1. cats
2. dogs
    a. zebras
    b. giraffes
        i. rats
        ii. mice
        iii. squirrels
        iv. rabbits
            i. bass
            ii. walleye

- eagles
- sparrows
    - pigeons
    - geese

$1 \text{ cats}\{\} \text{ in the house} = 1$

$1 \text{ cats}\{ \text{ in the house} = 1$

1 cats} in the house = 1

2 dogs[] in the house = 2

2 dogs[in the house = 2

2 dogs] in the house = 2

3 zebras in the house = 3

4 giraffes! in the house = 4

5 mice% in the house = 5

6 squirrels& in the house = 6

7 rats^ in the house = 7

8 horses# in the house = 8

9 cows @ in the house = 9

10 pigs $ in the house = 10

11 goats * in the house = 11

12 rabbits() in the house = 12

12 rabbits) in the house = 12

12 rabbits(in the house = 12

13 sparrow -+/<>.,';:~?<> = 13

14 fox _ in the house = 14

15 bear\ in the house = 15

Link Colors

Link test www.cnn.com

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